

Week 6 Completion Status:

- Established a reproducible, trace-driven simulation harness (testbed) to serve as the training environment for the reinforcement learning agent. (
- Integrated the Linear Regression forecasting model (from Week 5) into the environment's observation state to establish the predictor-to-policy loop.
- Implemented standard experiment configurations (fixed random seeds, bounded action spaces) to ensure reproducibility.
- **Updated git:** <https://github.com/scr200/CP395-Research-Project> (added week 6 simulation code to src folder)

Methodology/Changes:

This week, I transitioned from data analysis to systems engineering by building a custom, trace-driven simulator (CloudAutoscalingEnv) that mimics a cloud cluster. I utilized the OpenAI Gym API standards to ensure future compatibility with standard reinforcement learning libraries.

- **Predictor -> Policy Loop Integration:** I embedded the trained Linear Regression model into the environment. At each time step, the environment provides the agent with an observation array [current_demand, current_replicas, forecast_demand], allowing the policy to make proactive decisions based on the immediate future.
- **Heuristic Baseline Agent:** To validate the testbed, I implemented a rule-based Heuristic Agent. This agent acts as a baseline policy: it automatically scales up (+1) if the forecast exceeds current capacity, scales down (-1) if there is excessive buffer, and does nothing (0) otherwise.
- **Key Experiment Parameters:** * Random Seed: 42 (for strict reproducibility)
 - Hardware Limits: MIN_REPLICAS = 1, MAX_REPLICAS = 100, CAPACITY_PER_NODE = 64 Cores.
 - Action Space: Discrete (-1, 0, +1) to mimic real-world Autoscaler damping.
 - Optimization Weights: COST_WEIGHT = 1, SLO_PENALTY_WEIGHT = 50

Results & Analysis:

To validate the harness, I executed the Heuristic Agent through the full 30-day "Bursty" workload trace. The simulation completed successfully, logging a total of **2,462 SLO Violations** across the timeframe.

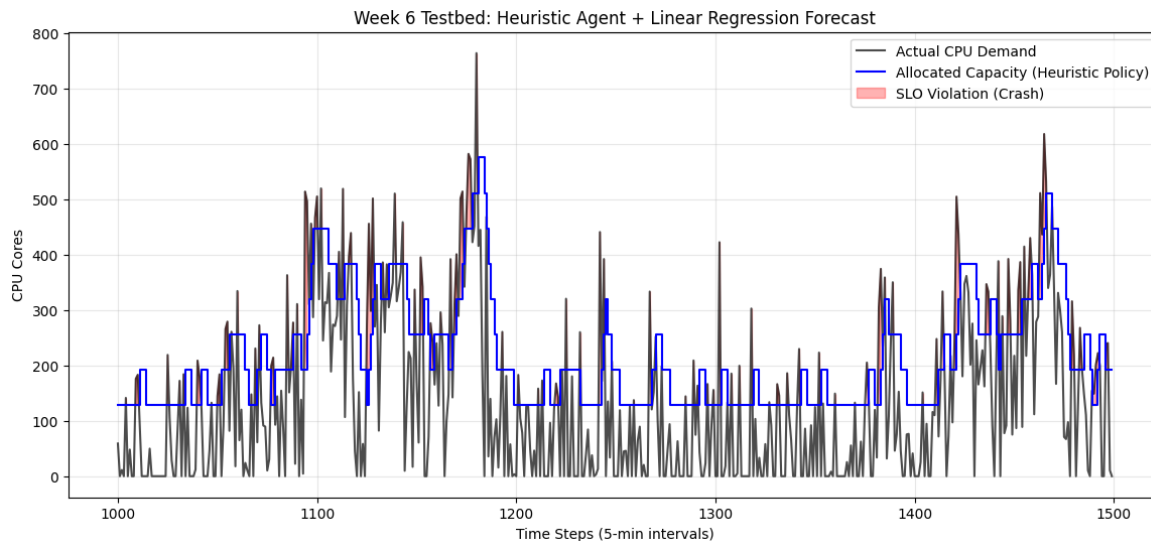


Figure 1: Week 6 Testbed running a Heuristic Policy against the Bursty Trace.

Interpretation: Figure 1 visually confirms that the simulator constraints are functioning perfectly. The Allocated Capacity (Blue Line) moves in strict step-wise increments, proving the +/- node scaling constraint is actively enforced. Furthermore, the Red Shaded areas represent SLO Violations (where Actual Demand > Capacity). This graph clearly illustrates the "Reaction Gap": even with a highly accurate Linear Regression forecast, a simple rule-based policy cannot scale fast enough to handle the severe non-linear spikes of a heavy-tailed workload. This formally establishes 2,462 violations as the baseline score that the future RL Agent must minimize.

Issues/Risks + Mitigation Actions:

- **Issue (Heuristic Failure):** The high number of violations (2,462) from the Heuristic Agent demonstrates that simply having a good forecast is insufficient; the *policy* mapping that forecast to an action must be highly sophisticated to handle risk.
- **Risk (Reward Tuning):** There is a risk that the Reinforcement Learning agent will also struggle to climb the demand spikes if the SLO_PENALTY_WEIGHT is not severe enough, leading it to prioritize cost over reliability.
- **Mitigation:** During RL training next week, I will treat the SLO_PENALTY_WEIGHT (currently 50) as a primary hyperparameter. If the agent under-provisions, I will exponentially increase this weight (e.g., to 100 or 500) to aggressively penalize crashes.

Next Steps for Week 6:

- **Algorithm Integration:** Install and integrate the Stable-Baselines3 library to utilize industry-standard RL algorithms.
- **Agent Training:** Instantiate a Proximal Policy Optimization (PPO) agent and train it within the CloudAutoscalingEnv simulator for 100,000+ timesteps.
- **Evaluation:** Run the fully trained PPO agent against the "Bursty" trace and compare its total SLO Violations against this week's heuristic baseline (2,462).